

# 1. Critical Rule: Frontal vs. Side View

You must identify the angle of the face before starting. This changes which regions you annotate.

## A. Frontal View (Facing Camera)

-  **ANNOTATE:** Forehead, Eyes, Nose, Mouth, Chin.
-  **DO NOT ANNOTATE:** Ears (p, b).

## B. Side Profile (Side View)

-  **ANNOTATE:** Only **Cheeks** and **Ears**.
-  **DO NOT ANNOTATE:** Eyes, Nose, Mouth, or Forehead.

---

# 2. Hotkey Cheat Sheet

| Key | Region Name                              | View  | Color Guide                                                                                      |
|-----|------------------------------------------|-------|--------------------------------------------------------------------------------------------------|
| 1   | Left Side Frontal                        | Front |  Green          |
| 2   | Right Side Frontal                       | Front |  Forest Green |
| 3   | Central Frontal                          | Front |  Blue         |
| 4   | Upper Eyelid                             | Front |  Yellow       |
| 5   | <b>Left</b> Lateral Canthus (Outer Eye)  | Front |  Magenta      |
| t   | <b>Right</b> Lateral Canthus (Outer Eye) | Front |  Deep Purple  |
| 6   | <b>Left</b> Medial Canthus (Inner Eye)   | Front |  Orange       |

|   |                                         |                                      |                                                                                                     |
|---|-----------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------|
| y | <b>Right</b> Medial Canthus (Inner Eye) | Front                                |  Goldenrod         |
| 7 | Mental Region (Chin)                    | Front                                |  Purple            |
| 8 | Border (Lip Outline / Lower Lip)        | Front                                |  Red               |
| 9 | Upper Lip                               | Front                                |  Cyan              |
| 0 | Nasal Region                            | Front                                |  Pinkish           |
| u | <b>Upper Cheek</b>                      | <b>SIDE<br/>&amp;<br/>FRO<br/>NT</b> |  Tomato            |
| l | <b>Lower Cheek</b>                      | <b>SIDE<br/>&amp;<br/>FRO<br/>NT</b> |  Cornflower Blue |
| p | <b>Ear Pinna</b> (Hard Cartilage)       | <b>SIDE</b>                          |  Teal            |
| b | <b>Ear Lobule</b> (Soft Lobe)           | <b>SIDE</b>                          |  Olive           |

## Controls

- **Left Click:** Add Point
- **Enter:** Finish Polygon (Close Shape)
- **s:** **Save** & Next Image
- **z:** **Undo** Last Polygon
- **x:** **Redo** Last Polygon
- **r:** **Reset** Current Drawing

- **q:** Quit / Skip Image
- 

## 3. Detailed Annotation Guidelines

### Frontal View Instructions

#### 1. Forehead (Keys 1, 2, 3)

- **3 (Central):** The middle vertical strip of the forehead (width of nose bridge).
- **1 (Left) / 2 (Right):** The remaining forehead areas on either side.
- **Boundary:** From Hairline (top) to Eyebrows (bottom).

#### 2. Eyes (Keys 4, 5, 6, t, y)

- **4 (Upper Eyelid):** Trace the skin fold *immediately* above the eyelashes.
- **5 / t (Lateral Canthus):** The small triangular skin patch at the **outer** corner of the eye ("Crow's feet").
  - Use **5** for the person's **Left** eye (Right side of screen).
  - Use **t** for the person's **Right** eye (Left side of screen).
- **6 / y (Medial Canthus):** The skin patch at the **inner** corner of the eye (near nose), including the tear duct.
  - Use **6** for Left Medial.
  - Use **y** for Right Medial.

#### 3. Cheeks (Keys u, l)

- **u (Upper Cheek):**
  - **Top:** Just below the eye.
  - **Bottom:** Imaginary horizontal line from the bottom of the nose to the ear.
- **l (Lower Cheek):**
  - **Top:** The imaginary line from the nose bottom.
  - **Bottom:** The Jawline.
  - **Front:** The Smile line (Nasolabial fold).

#### 3. Nose & Mouth (Keys 0, 8, 9, 7)

- **0 (Nasal):** The whole nose (bridge, tip, nostrils).
  - **9 (Upper Lip):** The top lip (fleshy part).
  - **8 (Border):** The bottom lip (fleshy part).
  - **7 (Chin):** The round part of the chin below the mouth (do not extend to jawline).
-

## Side Profile Instructions

### 1. Cheeks (Keys u, l)

- **u (Upper Cheek):**
  - **Top:** Just below the eye.
  - **Bottom:** Imaginary horizontal line from the bottom of the nose to the ear.
- **l (Lower Cheek):**
  - **Top:** The imaginary line from the nose bottom.
  - **Bottom:** The Jawline.
  - **Front:** The Smile line (Nasolabial fold).

### 2. Ears (Keys p, b)

- **p (Pinna):** The hard, upper cartilage part of the ear.
- **b (Lobule):** The soft, fleshy lobe at the bottom.

---

## 4. How to Annotate (Step-by-Step)

1. **Launch Script:** Run `python annotate.py`.
2. **Check Angle:** Is the person facing forward or sideways?
3. **Select Class:** Press the key for the region (e.g., `0` for Nose).
4. **Trace:** Click points along the edge. Use more points for curves (eyes), fewer for straight lines.
5. **Finish:** Press **Enter** to close the shape.
6. **Next Class:** Press the next key (e.g., `9` for Upper Lip) and repeat.
7. **Save:** When all visible regions for that view are done, press **s**.

## 5. Common Mistakes to Avoid

- **✗ Don't** annotate ears in frontal view.
- **✗ Don't** mix up Left and Right. Remember: "**Left**" means the person's **Left hand side**, not your left side.
- **✗ Don't** include hair in forehead or ear regions. Trace around it.